

MERU, Kenya

WMO: 636950

Lat: 0.083N

Lon: 37.650E

Elev: 1554

StdP: 84.00

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	11.7	7.9	8.0	16.6	9.0	8.6	16.0	7.6	19.5	6.4	19.5	1.4	250	0.274

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	11.2	27.2	18.0	26.3	18.0	25.6	17.9	20.1	23.9	19.7	23.4	19.4	23.1	2.6	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.0	16.7	22.2	18.6	16.2	21.8	18.2	15.8	21.4	65.7	24.0	64.1	23.5	62.9	23.2	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.9	4.7	4.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			7.7	22.2	1.3	1.4	6.8	23.2	6.0	24.0	5.3	24.8	4.4	25.8 (4)		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.9	18.5	19.5	20.2	19.9	19.2	18.0	17.4	17.9	19.1	19.9	19.0	18.3		
	DBStd	1.39	1.02	1.07	1.17	1.15	0.91	1.01	1.16	1.36	1.22	1.13	0.94	0.92		
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD18.3	115	9	2	1	1	3	17	34	24	7	2	4	11		
	CDD10.0	3243	263	266	316	296	284	241	228	244	272	306	269	258		
	CDD18.3	317	14	34	58	47	29	8	4	10	28	49	23	11		
	CDH23.3	1159	133	277	334	114	22	4	5	19	92	121	15	23		
(13)	CDH26.7	49	3	16	21	6	0	0	0	1	1	2	0	0		
(14)	Wind	WSAvg	1.8	1.5	1.6	1.6	2.0	1.9	2.1	2.3	2.3	1.8	1.2	1.2		
(15) Precipitation	PrecAvg	1371	52	39	117	285	121	9	8	11	18	251	317	144		
	PrecMax	2950	264	227	526	553	471	44	23	46	107	1387	889	439		
	PrecMin	669	0	0	2	78	9	0	0	0	0	2	97	0		
	PrecStd	497	57	56	120	120	94	11	6	11	22	242	166	114		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	28.2	27.3	24.8	24.0	24.0	24.0	24.9	26.3	27.0	24.6	25.2		
		MCWB	17.1	17.7	17.9	19.1	19.0	17.5	17.1	16.6	16.8	17.8	18.3	17.9		
	2%	DB	25.7	27.0	27.2	25.8	23.9	23.0	22.7	23.6	25.2	25.7	23.8	24.0		
		MCWB	17.6	17.9	18.2	18.7	18.8	17.4	16.4	16.2	16.7	17.5	18.4	18.1		
	5%	DB	24.8	26.1	26.2	24.6	23.2	22.2	21.9	22.7	24.4	24.7	23.1	23.2		
		MCWB	17.8	17.9	18.3	18.8	18.5	17.2	16.2	16.0	16.5	17.4	18.5	18.2		
	10%	DB	23.9	25.1	25.3	23.7	22.5	21.5	20.9	21.7	23.5	23.8	22.4	22.5		
		MCWB	17.9	18.0	18.3	18.8	18.4	17.0	15.9	15.8	16.3	17.4	18.4	18.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.2	20.2	21.8	20.6	20.3	19.0	18.1	17.9	18.3	19.6	19.8	20.1		
		MCDB	24.0	24.7	25.2	24.4	23.4	22.0	22.4	22.8	23.6	23.5	22.7	23.2		
	2%	WB	19.6	19.6	19.8	20.0	19.6	18.2	17.2	17.0	17.6	19.0	19.3	19.5		
		MCDB	23.5	24.5	24.4	23.5	22.7	21.6	21.3	21.8	23.3	22.8	22.4	22.5		
	5%	WB	19.1	19.1	19.3	19.6	19.2	17.7	16.7	16.5	17.1	18.5	19.0	19.1		
		MCDB	23.1	24.2	24.0	23.1	22.2	21.2	20.6	21.2	22.8	22.4	22.1	22.0		
	10%	WB	18.6	18.6	18.8	19.2	18.7	17.2	16.2	16.1	16.7	18.1	18.6	18.6		
		MCDB	22.6	23.8	23.7	22.7	21.7	20.5	19.9	20.6	22.1	22.1	21.6	21.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.7	12.6	11.2	8.5	7.6	7.5	7.4	8.1	9.6	8.7	7.7	9.0		
		MCDBR	13.5	14.0	13.0	10.0	8.6	8.8	9.2	9.9	11.0	10.2	8.7	10.5		
		MCWBR	7.2	6.8	5.7	4.6	4.4	4.2	4.0	3.9	4.1	3.7	4.5	5.8		
	5% WB	MCDBR	11.3	12.2	10.5	8.5	7.7	7.9	8.1	8.9	10.1	8.3	8.0	8.8		
		MCWBR	7.0	6.6	5.5	4.7	4.4	4.2	3.9	3.9	4.0	3.8	4.5	5.5		
(40) Clear-Sky Solar Irradiance	taub		0.349	0.365	0.363	0.338	0.333	0.385	0.394	0.366	0.343	0.340	0.347	0.356		
	taud		2.511	2.451	2.480	2.605	2.616	2.413	2.367	2.449	2.510	2.553	2.551	2.513		
	Ebn at Noon		980	965	959	962	944	880	877	922	963	977	975	969		
	Edh at Noon		111	119	115	99	94	113	120	114	110	106	106	109		
(44) All-Sky Solar Radiation	RadAvg		6.34	6.69	6.51	5.95	5.86	5.30	5.10	5.44	6.29	6.12	5.55	5.79		
	RadStd		0.41	0.28	0.43	0.30	0.17	0.27	0.34	0.27	0.28	0.23	0.26	0.34		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	+0.47	+0.69	N/A	+0.57	+0.39	+0.16	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page